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FINAL YEAR PROJECT REPORT

A Fractional-Order Compartmental Model of HIV Transmission with Social Behaviour Interventions

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Declaration

I, NDACYAYISABA Lambert, declare that this final year project report is my original work and has not been submitted elsewhere for any academic award. All sources used in this work have been acknowledged through proper citation and referencing.

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Approval

This final year project report has been submitted for examination with the approval of the university supervisor.

Supervisor: Dr. MUHIRWA Jean Pierre

Signature: ______________________________ **Date:** __________________

Dedication

This work is dedicated to my family, mentors, lecturers, and all people who believe that mathematics can be used to understand and improve human life.

Acknowledgements

I thank God for strength and guidance throughout this academic journey. I extend my gratitude to my supervisor, Dr. MUHIRWA Jean Pierre, for his guidance and valuable academic direction. I also appreciate the University of Rwanda, College of Science and Technology, and the Department of Applied Mathematics for providing the academic foundation for this work. I am grateful to my family, classmates, and friends for their support and encouragement.

Abstract

HIV/AIDS remains a major public health concern, particularly in sub-Saharan Africa, where transmission and control are shaped by both biomedical and social factors. Although antiretroviral therapy has improved survival and reduced transmission risk, HIV dynamics continue to be influenced by awareness, safer sexual behaviour, HIV testing, treatment-seeking behaviour, stigma, and adherence to medication. These factors suggest that HIV transmission cannot be fully understood through biological processes alone.

This report uses a fractional-order SITA model to analyse HIV transmission with social behaviour interventions. The population is divided into four compartments: susceptible individuals $S(t)$, infected individuals $I(t)$, treated individuals $T(t)$, and individuals in the AIDS stage $A(t)$. The model is formulated using the Caputo fractional derivative of order q, where $0 < q \leq 1$. The fractional order introduces memory effects, allowing the model to reflect the fact that behavioural change, treatment adherence, and public health interventions may influence transmission over time rather than instantaneously.

Special functions, especially the Gamma function and the Mittag-Leffler function, are introduced as essential tools in the formulation and analysis of the fractional model. The Gamma function appears in the definition of the Caputo derivative, while the Mittag-Leffler function plays a role similar to the exponential function in fractional dynamical systems. The report presents positivity, boundedness, disease-free equilibrium, the basic reproduction number $\mathcal{R}_0$, and local stability using fractional-order stability conditions. Numerical simulations are implemented using a fractional Adams–Bashforth–Moulton predictor-corrector method, and an interactive Python dashboard is used to visualize memory effects, intervention scenarios, sensitivity analysis, and epidemic trajectories.

The baseline simulation produced $\mathcal{R}_0 = 0.430 < 1$, indicating controlled transmission under the selected intervention-adjusted parameters. The fractional-order comparison showed that smaller values of q slow the decline of infected, treated, and AIDS-stage populations, while combined intervention scenarios produced the strongest reduction in infection.

Core Contribution

This report extends a standard HIV compartmental model by replacing ordinary derivatives with Caputo fractional derivatives and by incorporating social behaviour interventions into transmission, treatment uptake, and progression mechanisms. The work connects fractional calculus, special functions, epidemiological modelling, and computational simulation.

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List of Abbreviations and Symbols

Symbol	Meaning
HIV	Human Immunodeficiency Virus
AIDS	Acquired Immunodeficiency Syndrome
ART	Antiretroviral Therapy
SITA	Susceptible–Infected–Treated–AIDS model
$S(t)$	Susceptible population at time t
$I(t)$	Infected untreated population at time t
$T(t)$	Treated population at time t
$A(t)$	AIDS-stage population at time t
$N(t)$	Total population at time t
q	Fractional order, $0 < q \leq 1$
${}^{C}D_t^q$	Caputo fractional derivative of order q
$\Gamma(z)$	Gamma function
$E_q(z)$	Mittag-Leffler function
$\mathcal{R}_0$	Basic reproduction number
$u_1(t)$	Awareness and education intervention
$u_2(t)$	Safer sexual behaviour / condom use intervention
$u_3(t)$	Testing and treatment-seeking intervention
$u_4(t)$	Treatment adherence intervention

Chapter 1

Introduction

1.1 Background and Motivation

HIV/AIDS remains a major public health problem whose transmission is shaped by biological, behavioural, and social factors. Treatment through ART has changed HIV from a highly fatal disease into a manageable chronic condition for many people, but HIV transmission continues to be affected by awareness, testing behaviour, disclosure, stigma, safer sexual practices, and adherence to treatment (WHO, 2025; UNAIDS, 2025).

Mathematical modelling provides a systematic way of understanding how HIV spreads and how interventions may influence long-term dynamics. Classical compartmental models are commonly formulated using ordinary differential equations. However, ordinary models assume that the rate of change depends only on the current state of the system. This assumption may be restrictive in HIV modelling because social behaviour and treatment response often carry memory: past awareness campaigns, previous adherence patterns, and earlier testing behaviour can continue to influence present transmission.

Fractional calculus provides a natural extension of ordinary calculus by allowing derivatives of non-integer order. A fractional-order model can capture memory and hereditary effects, making it suitable for diseases and interventions whose dynamics are influenced by past states (Podlubny, 1999; Kilbas et al., 2006; Diethelm, 2010). For this reason, this report formulates a fractional-order HIV transmission model using the Caputo fractional derivative.

1.2 Rwanda and Public Health Context

Rwanda has made important progress in HIV prevention, testing, treatment, and care. The Rwanda Population-based HIV Impact Assessment 2018–2019 reported adult HIV prevalence of approximately 3.0% among adults aged 15–64 years, annual adult incidence around 0.08%, and viral load suppression among HIV-positive adults around 76.0% (RBC, 2020). These findings show progress but also indicate that HIV transmission has not been fully eliminated.

Social behaviour remains relevant in the Rwandan HIV response. Stigma, testing behaviour, disclosure, and adherence can affect prevention and treatment pathways (RRP+, 2020). Therefore, a model that combines fractional dynamics with social behaviour interventions can provide useful academic insight into how memory and behavioural change may influence HIV transmission.

1.3 Statement of the Problem

Many existing HIV models use ordinary differential equations and treat transmission as a direct function of current infected populations. Such models may not fully capture memory effects arising from past behaviour, treatment adherence, community awareness, or stigma reduction. In addition, social behaviour is often represented as a fixed parameter rather than as intervention functions that modify transmission, treatment uptake, and progression.

This report addresses this limitation by developing a fractional-order SITA model that incorporates social behaviour interventions. The model uses the Caputo derivative to include memory effects and introduces intervention functions representing awareness, safer sexual behaviour, testing and treatment-seeking, and treatment adherence.

The central mathematical problem is therefore to connect three components in one coherent framework: a biologically meaningful HIV compartmental model, a fractional-order derivative that represents memory, and intervention functions that can be varied computationally. This connection is important because it allows the model to be analysed theoretically and then explored numerically through a dashboard.

1.4 Aim of the Study

The main aim of this report is to formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions.

1.5 Specific Objectives

1. To formulate a fractional-order SITA HIV transmission model using the Caputo derivative and social behaviour intervention functions.
2. To analyse the main mathematical properties of the model, including positivity, boundedness, disease-free equilibrium, $\mathcal{R}_0$, and fractional-order stability.
3. To implement numerical simulations and a Python dashboard for studying memory effects, intervention scenarios, sensitivity, and epidemic trajectories.

1.6 Research Questions

1. How can a SITA HIV transmission model be formulated using the Caputo fractional derivative and social behaviour interventions?
2. What are the main analytical properties of the model, including positivity, boundedness, $\mathcal{R}_0$, and local stability?
3. How do fractional memory and intervention scenarios affect the simulated HIV dynamics in the dashboard?

1.7 Significance of the Study

This study is significant in three ways. Mathematically, it introduces fractional calculus and special functions into HIV modelling and shows how the Caputo derivative, Gamma function, and Mittag-Leffler function enter the analysis of a biological system. Epidemiologically, it provides a structured framework for comparing behavioural and treatment-related interventions through the basic reproduction number and numerical trajectories. Computationally, it supports the development of an interactive dashboard that makes the fractional model testable, reproducible, and easier to communicate during presentation.

The work is also relevant to applied mathematics because it combines model formulation, qualitative analysis, numerical approximation, parameter-based simulation, and software implementation. These components show how mathematical theory can be translated into an exploratory tool for understanding disease dynamics.

1.8 Scope of the Study

The study focuses on a deterministic fractional-order SITA model with four compartments. It considers social behaviour interventions but does not include age structure, gender structure, spatial movement, stochasticity, or direct patient-level data. The dashboard is used for academic simulation and interpretation, not clinical prediction or direct policy forecasting.

All numerical results should therefore be interpreted as scenario-based mathematical experiments. The purpose is to compare the qualitative effects of memory and intervention levels, not to estimate exact HIV prevalence or incidence in a real population.

Chapter 2

Literature Review and Mathematical Preliminaries

2.1 Compartmental Models in Epidemiology

Compartmental epidemic modelling has its roots in the work of Kermack and McKendrick (1927), who introduced the classical SIR framework. Later studies extended such models to represent different infectious disease dynamics. Hethcote (2000) provides a comprehensive review of infectious disease models, including threshold quantities such as the basic reproduction number.

2.2 HIV/AIDS Mathematical Models

HIV differs from many acute infections because infected individuals do not naturally move into a permanently recovered class. For this reason, HIV models often include treatment and AIDS progression compartments (Anderson and May, 1991; Silva and Torres, 2017). In this report, the population is divided into susceptible, infected, treated, and AIDS classes.

Treatment compartments are important because antiretroviral therapy changes both the clinical progression of HIV and the effective contribution of treated individuals to transmission. Models such as SICA or SITA-type systems are therefore useful because they separate untreated infection from treatment and advanced disease stages. This separation allows the modeller to study treatment uptake, progression to AIDS, and disease-induced mortality within the same framework.

2.3 Social Behaviour in HIV Transmission

HIV transmission is strongly influenced by behavioural and social factors. These include condom use, number of sexual partners, HIV testing, disclosure, stigma, treatment-seeking behaviour, and adherence to ART. Mathematical models can represent these effects through risk groups, behaviour-dependent transmission rates, or intervention functions (Poundstone et al., 2004; Supervie et al., 2011).

Social behaviour is especially relevant in HIV modelling because infection risk is not determined only by biological infectiousness. Awareness campaigns, safer sexual behaviour, willingness to test, and adherence support can change the effective contact process and the movement of infected individuals into treatment. In the present report, these factors are represented by bounded intervention functions so that their separate and combined effects can be compared numerically.

2.4 Fractional Calculus in Epidemiological Models

Fractional calculus extends ordinary differentiation and integration to non-integer orders. In epidemiological modelling, fractional derivatives are useful because they can incorporate memory and hereditary effects (Podlubny, 1999; Kilbas et al., 2006; Diethelm, 2010). Several fractional HIV/AIDS models have been proposed using Caputo and related operators (Günerhan et al., 2020; Unaegbu et al., 2021).

The motivation for using fractional derivatives is that epidemic processes may depend on past states rather than only on the current state. For HIV, this is meaningful because public-health education, treatment adherence, stigma reduction, and testing habits often influence behaviour over a period of time. A Caputo fractional model therefore provides a mathematical way to include delayed and persistent effects while still using standard initial conditions.

Numerically, fractional models require methods that account for the memory integral. Predictor–corrector methods of Adams–Bashforth–Moulton type are commonly used for Caputo systems because each new time step depends on a weighted history of earlier function values (Diethelm et al., 2002; Diethelm, 2010). This is the numerical approach adopted in the computational part of the report.

2.5 Gamma Function

The Gamma function is a fundamental special function in fractional calculus. It generalizes the factorial function and is defined by

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad z > 0. \tag{2.1}$$

For positive integers n, it satisfies

$$\Gamma(n) = (n-1)!. \tag{2.2}$$

The Gamma function appears in the definition of fractional integrals and derivatives.

2.6 Riemann–Liouville Fractional Integral

For a function f and order $q > 0$, the Riemann–Liouville fractional integral is defined as

$$I_t^q f(t) = \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} f(s)\, ds. \tag{2.3}$$

This integral operator shows explicitly how past values of f contribute to the present value, weighted by a power-law memory kernel.

2.7 Caputo Fractional Derivative

For $0 < q < 1$, the Caputo fractional derivative of a differentiable function f is defined by

$$^{C}D_t^q f(t) = \frac{1}{\Gamma(1-q)} \int_0^t \frac{f'(s)}{(t-s)^q}\, ds. \tag{2.4}$$

The Caputo derivative is used in this report because it allows classical initial conditions, such as $S(0) = S_0$, $I(0) = I_0$, $T(0) = T_0$, and $A(0) = A_0$ (Caputo, 1967; Diethelm, 2010).

Why the Caputo Derivative is Selected

The Caputo derivative is selected because it is suitable for biological models with standard initial conditions. It also carries a memory kernel, meaning that the present dynamics depend on past states. This is important for HIV modelling because awareness, adherence, stigma reduction, and testing behaviour may influence transmission over time rather than only at the present moment.

2.8 Mittag-Leffler Function

The Mittag-Leffler function is a special function that generalizes the exponential function in fractional calculus. The one-parameter Mittag-Leffler function is defined by

$$E_q(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(qk+1)}, \qquad q > 0. \tag{2.5}$$

The two-parameter form is

$$E_{q,r}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(qk+r)}, \qquad q, r > 0. \tag{2.6}$$

When $q = 1$, the Mittag-Leffler function reduces to the exponential function:

$$E_1(z) = e^z. \tag{2.7}$$

This makes the Mittag-Leffler function central in fractional differential equations, where it often replaces the exponential function in solutions and stability analysis (Kilbas et al., 2006; Diethelm, 2010).

2.9 Fractional Stability Criterion

For a linear fractional system

$$^{C}D_t^q x(t) = Ax(t), \qquad 0 < q \leq 1, \tag{2.8}$$

the equilibrium is locally asymptotically stable if every eigenvalue λ_i of A satisfies

$$|\arg(\lambda_i)| > \frac{q\pi}{2}. \tag{2.9}$$

This criterion is more general than the ordinary condition $\mathrm{Re}(\lambda_i) < 0$ and becomes the classical condition when $q = 1$ (Matignon, 1996).

Chapter 3

Fractional-Order Model Formulation

3.1 Population Compartments

The total population at time t is divided into four compartments:

Compartment	Description
$S(t)$	Susceptible individuals who are not infected but are at risk of HIV infection.
$I(t)$	Infected individuals not yet on treatment.
$T(t)$	HIV-positive individuals receiving ART.
$A(t)$	Individuals in the AIDS stage.

The total population is

$$N(t) = S(t) + I(t) + T(t) + A(t). \tag{3.1}$$

3.2 Model Assumptions

The model is formulated under the following assumptions:

1. The population is divided into four mutually exclusive compartments: susceptible, infected, treated, and AIDS.

2. Individuals enter the susceptible class through recruitment at rate Λ.

3. Susceptible individuals acquire HIV through effective contact with infected or treated individuals.

4. Treated individuals have reduced infectiousness compared with untreated infected individuals, represented by η, where $0 < \eta < 1$.

5. Infected individuals may begin treatment and move into the treated class.

6. Infected individuals may progress to AIDS at rate δ.

7. Treated individuals may progress to AIDS at rate ρ, but adherence intervention reduces this progression.

8. Natural mortality occurs in all compartments at rate μ.

9. AIDS-induced mortality occurs in the AIDS compartment at rate d.

10. Social behaviour interventions are bounded controls satisfying $0 \leq u_i(t) \leq 1$, $i = 1, 2, 3, 4$.

11. The fractional order satisfies $0 < q \leq 1$, where $q = 1$ gives the classical ordinary model and $0 < q < 1$ introduces memory.

12. During the analytical derivation of $\mathcal{R}_0$, the intervention levels are treated as constants so that the disease-free equilibrium and next-generation matrix can be defined clearly.

13. The model is intended for qualitative and comparative analysis; parameter values used in simulations represent baseline academic scenarios rather than direct clinical calibration.

These assumptions simplify the complex social and biological processes involved in HIV transmission. They are useful for constructing a mathematically tractable model, but they also define the limitations of the conclusions drawn from the simulations.

3.3 Social Behaviour Intervention Functions

The model includes four social behaviour interventions:

Intervention	Meaning	Range
$u_1(t)$	Awareness and education intervention	$0 \leq u_1(t) \leq 1$
$u_2(t)$	Condom use / safer sexual behaviour intervention	$0 \leq u_2(t) \leq 1$
$u_3(t)$	Testing and treatment-seeking intervention	$0 \leq u_3(t) \leq 1$
$u_4(t)$	Treatment adherence intervention	$0 \leq u_4(t) \leq 1$

3.4 Model Flow Diagram

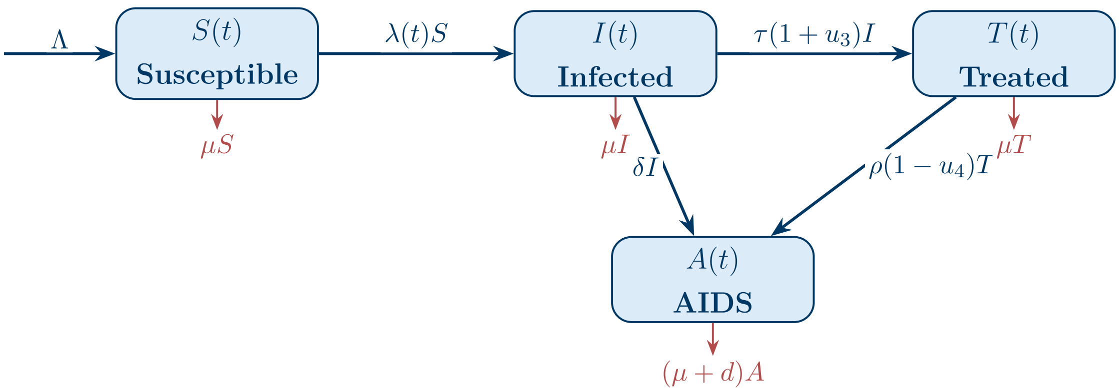


Figure 3.1: Compartmental flow diagram for the SITA HIV model. The figure is placed before the equations to show the population transfers before the mathematical formulation.

3.5 Effective Transmission and Transition Terms

The effective transmission rate is defined as

$$\beta_{\text{eff}}(t) = \beta_0(1 - u_1(t))(1 - u_2(t)). \tag{3.2}$$

The force of infection is

$$\lambda(t) = \frac{\beta_{\text{eff}}(t)(I + \eta T)}{N}, \tag{3.3}$$

where $0 < \eta < 1$ represents the reduced infectiousness of treated individuals.

The effective treatment uptake and treated-to-AIDS progression rates are

$$\tau_{\text{eff}}(t) = \tau(1 + u_3(t)), \tag{3.4}$$

and

$$\rho_{\text{eff}}(t) = \rho(1 - u_4(t)). \tag{3.5}$$

3.6 Fractional-Order SITA Model

Using the Caputo fractional derivative of order q, $0 < q \leq 1$, the proposed model is

$$^{C}D_t^q S(t) = \Lambda - \lambda(t)S - \mu S, \tag{3.6}$$

$$^{C}D_t^q I(t) = \lambda(t)S - [\tau(1 + u_3(t)) + \delta + \mu]I, \tag{3.7}$$

$$^{C}D_t^q T(t) = \tau(1 + u_3(t))I - [\rho(1 - u_4(t)) + \mu]T, \tag{3.8}$$

$$^{C}D_t^q A(t) = \delta I + \rho(1 - u_4(t))T - (\mu + d)A. \tag{3.9}$$

The initial conditions are

$$S(0) = S_0, \qquad I(0) = I_0, \qquad T(0) = T_0, \qquad A(0) = A_0. \tag{3.10}$$

Connection with the Ordinary Model

When $q = 1$, the Caputo derivative becomes the ordinary first derivative and the fractional-order model reduces to the classical SITA ordinary differential equation model. When $0 < q < 1$, the model includes memory effects.

3.7 Parameter Description

Parameter	Description	Range
Λ	Recruitment rate into susceptible class	> 0
μ	Natural mortality rate	> 0
β_0	Baseline HIV transmission rate	> 0
η	Relative infectiousness of treated individuals	$0 < \eta < 1$
τ	Baseline treatment initiation rate	> 0
δ	Progression rate from infected class to AIDS	> 0
ρ	Progression rate from treated class to AIDS	> 0
d	AIDS-induced mortality rate	> 0
q	Fractional order representing memory effect	$0 < q \leq 1$
$u_1(t)$	Awareness and education intervention	$0 \leq u_1 \leq 1$
$u_2(t)$	Safer sexual behaviour intervention	$0 \leq u_2 \leq 1$
$u_3(t)$	Testing and treatment-seeking intervention	$0 \leq u_3 \leq 1$
$u_4(t)$	Treatment adherence intervention	$0 \leq u_4 \leq 1$

The parameter values used in the simulations are selected from three sources: demographic and HIV context reports, published HIV modelling studies, and normalized values chosen for controlled scenario comparison. Rwanda-specific HIV context is guided by RPHIA and public-health reports (RBC, 2020; RRP+, 2020), while the compartmental HIV structure and transition parameters are guided by standard HIV/AIDS modelling literature (Anderson and May, 1991; Silva and Torres, 2017; Günerhan et al., 2020; Unaegbu et al., 2021). The dashboard therefore uses literature-informed baseline values for academic simulation rather than claiming direct calibration to patient-level Rwanda data.

Parameter	Role in the model	Source or modelling basis
Λ	Recruitment into the susceptible population	Normalized simulation scale
μ	Natural mortality rate	Demographic modelling context
β_0	Baseline transmission intensity	HIV model literature
η	Relative infectiousness under treatment	ART-reduced infectiousness
τ	Treatment initiation rate	HIV treatment-model literature
δ	Progression from infection to AIDS	HIV progression literature
ρ	Progression from treatment to AIDS	Treatment progression literature
d	AIDS-induced mortality	HIV/AIDS model literature
q	Fractional memory order	Fractional model scenario
u_1, u_2, u_3, u_4	Social behaviour intervention levels	Intervention scenarios

Chapter 4

Model Analysis

Throughout this chapter, the intervention functions are assumed to satisfy

$$0 \leq u_i(t) \leq 1, \qquad i = 1, 2, 3, 4,$$

and all biological parameters are assumed to be nonnegative, with

$$\Lambda > 0, \quad \mu > 0, \quad \beta_0 > 0, \quad \tau > 0, \quad \delta > 0, \quad \rho > 0, \quad d > 0, \quad 0 < \eta < 1.$$

The fractional order satisfies

$$0 < q \leq 1.$$

For mathematical analysis, the system is written in compact form as

$${}^{C}D_t^q X(t) = F(t, X(t)), \qquad X(t) = \big(S(t), I(t), T(t), A(t)\big)^T.$$

4.1 Equivalent Integral Form

The Caputo fractional system can be rewritten as an equivalent Volterra integral system by applying the Riemann–Liouville fractional integral. This representation is important because fractional differential equations are naturally nonlocal, meaning that the present state depends on the entire previous history of the system.

For the susceptible compartment, the equivalent integral form is

$$S(t) = S_0 + \frac{1}{\Gamma(q)} \int_0^t (t - s)^{q-1} \left[\Lambda - \lambda(s)S(s) - \mu S(s)\right] ds. \tag{4.1}$$

Similarly, for the infected, treated, and AIDS-stage compartments, we obtain

$$I(t) = I_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} \Big[\lambda(s)S(s) - \Big(\tau(1+u_3(s)) + \delta + \mu\Big)I(s)\Big]\, ds, \tag{4.2}$$

$$T(t) = T_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} \Big[\tau(1+u_3(s))I(s) - \Big(\rho(1-u_4(s)) + \mu\Big)T(s)\Big]\, ds, \tag{4.3}$$

$$A(t) = A_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} \left[\delta I(s) + \rho(1-u_4(s))T(s) - (\mu + d)A(s)\right] ds. \tag{4.4}$$

4.2 Existence and Uniqueness of Solutions

For a mathematical model to be meaningful, it must have a solution and that solution should be unique under the stated initial conditions. Let

$$\Omega_M = \Big\{X \in \mathbb{R}^4_+ : 0 \le S + I + T + A \le M,\ S + I + T + A > 0\Big\}$$

be a bounded positive region for some $M > 0$. In this region, the force of infection

$$\lambda(t) = \frac{\beta_0(1-u_1(t))(1-u_2(t))(I+\eta T)}{N}$$

is continuous whenever $N > 0$.

Theorem 4.1 — Existence and Uniqueness

Suppose that the initial data satisfy

$$S_0 \ge 0, \qquad I_0 \ge 0, \qquad T_0 \ge 0, \qquad A_0 \ge 0, \qquad N_0 = S_0 + I_0 + T_0 + A_0 > 0.$$

If the intervention functions $u_i(t)$, $i = 1, 2, 3, 4$, are bounded and continuous on $[0, T]$, then the fractional-order SITA system admits a unique local solution on a suitable interval $[0, T^*] \subseteq [0, T]$.

Proof. Using the Caputo integral representation, the system can be written as

$$X(t) = X(0) + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} F(s, X(s)) ds.$$

Define the operator

$$(\mathcal{T}X)(t) = X(0) + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} F(s, X(s)) ds.$$

On any bounded subset of Ω_M, the function $F(t, X)$ is continuous in t and locally Lipschitz in X, because it is made from sums, products, and rational terms whose denominator N is positive. Hence, there exists a constant $L > 0$ such that

$$\|F(t, X) - F(t, Y)\| \leq L\|X - Y\|.$$

For $X, Y \in C([0, T^*], \mathbb{R}^4)$, we have

$$\|\mathcal{T}X - \mathcal{T}Y\| \leq \frac{L}{\Gamma(q)} \int_0^{T^*} (T^* - s)^{q-1} \|X - Y\| ds.$$

Therefore,

$$\|\mathcal{T}X - \mathcal{T}Y\| \leq \frac{L(T^*)^q}{\Gamma(q+1)} \|X - Y\|.$$

Choosing $T^* > 0$ sufficiently small so that

$$\frac{L(T^*)^q}{\Gamma(q+1)} < 1,$$

the operator $\mathcal{T}$ becomes a contraction. By Banach's fixed point theorem, the model has a unique local solution. Since boundedness is established below, the solution can be extended while it remains in the biologically feasible region. □

4.3 Positivity of Solutions

Population variables cannot be negative. Thus, positivity is an essential biological requirement.

Theorem 4.2 — Positivity

If

$$S_0 \geq 0, \qquad I_0 \geq 0, \qquad T_0 \geq 0, \qquad A_0 \geq 0,$$

then the solution of the fractional-order SITA model satisfies

$$S(t) \geq 0, \qquad I(t) \geq 0, \qquad T(t) \geq 0, \qquad A(t) \geq 0, \qquad t \geq 0.$$

Proof. We verify that the vector field points inward on the boundary of the nonnegative orthant. At $S = 0$,

$${}^C D_t^q S(t) = \Lambda \geq 0.$$

At $I = 0$,

$${}^C D_t^q I(t) = \lambda(t) S(t) \geq 0.$$

At $T = 0$,

$$^{C}D_t^q T(t) = \tau(1 + u_3(t))I(t) \geq 0.$$

At $A = 0$,

$$^{C}D_t^q A(t) = \delta I(t) + \rho(1 - u_4(t))T(t) \geq 0.$$

Since each boundary component has a nonnegative fractional derivative in the corresponding compartment, the nonnegative orthant is positively invariant by the standard positivity principle for Caputo fractional systems. Therefore, solutions starting with nonnegative initial conditions remain nonnegative for all $t \geq 0$. $\square$

4.4 Boundedness and Feasible Region

Boundedness ensures that the total population remains finite. Let

$$N(t) = S(t) + I(t) + T(t) + A(t).$$

Adding the four model equations gives

$$^{C}D_t^q N(t) = \Lambda - \mu N(t) - dA(t). \tag{4.5}$$

Since $dA(t) \geq 0$, it follows that

$$^{C}D_t^q N(t) \leq \Lambda - \mu N(t). \tag{4.6}$$

Consider the comparison equation

$$^{C}D_t^q Y(t) = \Lambda - \mu Y(t), \qquad Y(0) = N(0).$$

Its solution is

$$Y(t) = \frac{\Lambda}{\mu} + \left(N(0) - \frac{\Lambda}{\mu}\right) E_q(-\mu t^q), \tag{4.7}$$

where $E_q(\cdot)$ is the Mittag–Leffler function.

Theorem 4.3 — Boundedness

All solutions of the fractional-order SITA model with nonnegative initial conditions are bounded. If

$$M_0 = \max\left\{N(0), \frac{\Lambda}{\mu}\right\},$$

then the solution remains in the finite-time feasible envelope

$$\Omega = \left\{(S, I, T, A) \in \mathbb{R}_+^4 : 0 \leq N(t) \leq M_0\right\}$$

and, in the long run,

$$\limsup_{t\to\infty} N(t) \leq \frac{\Lambda}{\mu}.$$

Proof. By the fractional comparison principle, inequality (4.6) implies

$$N(t) \leq Y(t),$$

where $Y(t)$ is given by (4.7). Since $E_q(-\mu t^q) \to 0$ as $t \to \infty$ for $\mu > 0$, we obtain

$$\limsup_{t\to\infty} N(t) \leq \frac{\Lambda}{\mu}.$$

Therefore, the total population remains bounded and the region Ω is biologically feasible. □

4.5 Disease-Free Equilibrium

The disease-free equilibrium represents the state where HIV is absent from the population. Therefore,

$$I = T = A = 0.$$

Substituting these values into the susceptible equation gives

$$0 = \Lambda - \mu S.$$

Hence,

$$E_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0\right). \tag{4.8}$$

4.6 Basic Reproduction Number

For the derivation of $\mathcal{R}_0$, the intervention functions are treated as constant levels:

$$u_i(t) = u_i, \qquad i = 1, 2, 3, 4.$$

Define

$$\beta_{\text{eff}} = \beta_0(1-u_1)(1-u_2), \qquad \tau_{\text{eff}} = \tau(1+u_3), \qquad \rho_{\text{eff}} = \rho(1-u_4).$$

At the disease-free equilibrium,

$$S^* = \frac{\Lambda}{\mu}, \qquad N^* = \frac{\Lambda}{\mu}.$$

Thus, near E_0,

$$\lambda(t)S = \beta_{\text{eff}}(I+\eta T).$$

The infected-related compartments are I and T. The new infection terms are

$$\mathcal{F}_1 = \beta_{\text{eff}}(I+\eta T), \qquad \mathcal{F}_2 = 0.$$

The transition terms are

$$\mathcal{V}_1 = (\tau_{\text{eff}}+\delta+\mu)I,$$

and

$$\mathcal{V}_2 = (\rho_{\text{eff}}+\mu)T - \tau_{\text{eff}}I.$$

Therefore,

$$F = \begin{pmatrix} \beta_{\text{eff}} & \eta\beta_{\text{eff}} \\ 0 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} \tau_{\text{eff}}+\delta+\mu & 0 \\ -\tau_{\text{eff}} & \rho_{\text{eff}}+\mu \end{pmatrix}.$$

The inverse of V is

$$V^{-1} = \frac{1}{(\tau_{\text{eff}}+\delta+\mu)(\rho_{\text{eff}}+\mu)} \begin{pmatrix} \rho_{\text{eff}}+\mu & 0 \\ \tau_{\text{eff}} & \tau_{\text{eff}}+\delta+\mu \end{pmatrix}.$$

Following the standard next-generation matrix approach (Diekmann et al., 1990; Van den Driessche and Watmough, 2002), the next-generation matrix is

$$K = FV^{-1}.$$

Multiplying gives

$$K = \frac{\beta_{\text{eff}}}{(\tau_{\text{eff}}+\delta+\mu)(\rho_{\text{eff}}+\mu)} \begin{pmatrix} \rho_{\text{eff}}+\mu+\eta\tau_{\text{eff}} & \eta(\tau_{\text{eff}}+\delta+\mu) \\ 0 & 0 \end{pmatrix}.$$

The basic reproduction number is the spectral radius of K. Hence,

$$\mathcal{R}_0 = \frac{\beta_{\text{eff}}}{\tau_{\text{eff}} + \delta + \mu}\left(1 + \frac{\eta\tau_{\text{eff}}}{\rho_{\text{eff}} + \mu}\right). \tag{4.9}$$

Substituting the intervention-modified rates gives

$$\mathcal{R}_0 = \frac{\beta_0(1-u_1)(1-u_2)}{\tau(1+u_3) + \delta + \mu}\left(1 + \frac{\eta\tau(1+u_3)}{\rho(1-u_4) + \mu}\right). \tag{4.10}$$

Important interpretation of u_4

In the present model, u_4 reduces progression from treatment to AIDS through $\rho(1-u_4)$. Since the treated class still contributes to transmission through ηT, the direct effect of u_4 on $\mathcal{R}_0$ can be subtle. If the study aims to make adherence reduce infectiousness directly, a refined force of infection may use $\eta(1-u_4)T$. In the current structure, u_4 is mainly interpreted as reducing AIDS progression and improving treatment outcomes, while u_1 and u_2 directly reduce new infections.

4.7 Local Stability of the Disease-Free Equilibrium

The Jacobian matrix of the full system at E_0 is

$$J(E_0) = \begin{pmatrix} -\mu & -\beta_{\text{eff}} & -\eta\beta_{\text{eff}} & 0 \\ 0 & \beta_{\text{eff}} - (\tau_{\text{eff}} + \delta + \mu) & \eta\beta_{\text{eff}} & 0 \\ 0 & \tau_{\text{eff}} & -(\rho_{\text{eff}} + \mu) & 0 \\ 0 & \delta & \rho_{\text{eff}} & -(\mu + d) \end{pmatrix}.$$

Two eigenvalues are immediately obtained:

$$\lambda_1 = -\mu, \qquad \lambda_2 = -(\mu + d),$$

which are both negative.

The remaining eigenvalues are obtained from the infected subsystem

$$J_{IT} = \begin{pmatrix} \beta_{\text{eff}} - (\tau_{\text{eff}} + \delta + \mu) & \eta\beta_{\text{eff}} \\ \tau_{\text{eff}} & -(\rho_{\text{eff}} + \mu) \end{pmatrix}.$$

Let

$$a = \tau_{\text{eff}} + \delta + \mu, \qquad b = \rho_{\text{eff}} + \mu.$$

Then

$$J_{IT} = \begin{pmatrix} \beta_{\text{eff}} - a & \eta\beta_{\text{eff}} \\ \tau_{\text{eff}} & -b \end{pmatrix}.$$

The trace is

$$\text{tr}(J_{IT}) = \beta_{\text{eff}} - a - b,$$

and the determinant is

$$\det(J_{IT}) = (\beta_{\text{eff}} - a)(-b) - \eta\beta_{\text{eff}}\tau_{\text{eff}}.$$

After simplification,

$$\det(J_{IT}) = ab - \beta_{\text{eff}}(b + \eta\tau_{\text{eff}}). \tag{4.11}$$

Using (4.9), we obtain

$$\det(J_{IT}) = ab(1 - \mathcal{R}_0). \tag{4.12}$$

Theorem 4.4 — Fractional Local Stability of E_0

The disease-free equilibrium

$$E_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0\right)$$

is locally asymptotically stable in the fractional-order sense if

$$\mathcal{R}_0 < 1$$

and the eigenvalues λ_i of the linearized system satisfy

$$|\arg(\lambda_i)| > \frac{q\pi}{2}.$$

It is unstable if $\mathcal{R}_0 > 1$.

Proof. If $\mathcal{R}_0 < 1$, then from (4.12),

$$\det(J_{IT}) = ab(1 - \mathcal{R}_0) > 0.$$

Also, $\mathcal{R}_0 < 1$ implies

$$\beta_{\text{eff}} < \frac{ab}{b + \eta\tau_{\text{eff}}} < a,$$

because $b + \eta\tau_{\text{eff}} > b$. Therefore,

$$\text{tr}(J_{IT}) = \beta_{\text{eff}} - a - b < 0.$$

Thus, the infected subsystem has eigenvalues with negative real parts. Since the remaining eigenvalues $-\mu$ and $-(\mu+d)$ are also negative, the linearized ordinary system is locally stable.

For a Caputo fractional system of order $0 < q \leq 1$, local asymptotic stability requires

$$|\arg(\lambda_i)| > \frac{q\pi}{2}.$$

Eigenvalues with negative real parts satisfy this condition for $0 < q \leq 1$. Hence E_0 is locally asymptotically stable in the fractional-order sense when $\mathcal{R}_0 < 1$.

If $\mathcal{R}_0 > 1$, then

$$\det(J_{IT}) = ab(1 - \mathcal{R}_0) < 0,$$

so the infected subsystem has eigenvalues with opposite signs in the real case or at least one eigenvalue with positive real part. Therefore, the disease-free equilibrium is unstable. □

4.8 Endemic Equilibrium

An endemic equilibrium represents a persistent infection state:

$$E^* = (S^*, I^*, T^*, A^*), \qquad I^* > 0, \quad T^* > 0, \quad A^* > 0.$$

At endemic equilibrium,

$${}^C D_t^q S = {}^C D_t^q I = {}^C D_t^q T = {}^C D_t^q A = 0.$$

Therefore, the equilibrium equations are

$$0 = \Lambda - \lambda^* S^* - \mu S^*, \tag{4.13}$$

$$0 = \lambda^* S^* - (\tau_{\text{eff}} + \delta + \mu) I^*, \tag{4.14}$$

$$0 = \tau_{\text{eff}} I^* - (\rho_{\text{eff}} + \mu) T^*, \tag{4.15}$$

$$0 = \delta I^* + \rho_{\text{eff}} T^* - (\mu + d) A^*. \tag{4.16}$$

From the third and fourth equations,

$$T^* = \frac{\tau_{\text{eff}}}{\rho_{\text{eff}} + \mu} I^*,$$

and

$$A^* = \frac{\delta I^* + \rho_{\text{eff}} T^*}{\mu + d}.$$

The explicit expression for I^* may be obtained by substituting these relations into the force of infection and susceptible equilibrium equation. Because the resulting expression is lengthy, this report uses numerical simulation to examine endemic behaviour when $\mathcal{R}_0 > 1$.

4.9 Sensitivity Analysis

The normalized sensitivity index of $\mathcal{R}_0$ with respect to a parameter p is

$$\Upsilon_p^{\mathcal{R}_0} = \frac{\partial \mathcal{R}_0}{\partial p} \cdot \frac{p}{\mathcal{R}_0}. \tag{4.17}$$

This index measures the relative change in $\mathcal{R}_0$ caused by a relative change in p.

From (4.10), the following direct sensitivity signs are obtained:

$$\Upsilon_{\beta_0}^{\mathcal{R}_0} = 1,$$

which means that a percentage increase in β_0 produces the same percentage increase in $\mathcal{R}_0$.

For the awareness and safer behaviour interventions,

$$\Upsilon_{u_1}^{\mathcal{R}_0} = -\frac{u_1}{1 - u_1}, \qquad \Upsilon_{u_2}^{\mathcal{R}_0} = -\frac{u_2}{1 - u_2}.$$

Thus, increasing u_1 or u_2 decreases $\mathcal{R}_0$, confirming that awareness and safer sexual behaviour directly reduce transmission.

For the AIDS progression parameter δ,

$$\Upsilon_{\delta}^{\mathcal{R}_0} = -\frac{\delta}{\tau_{\text{eff}} + \delta + \mu}.$$

Hence increasing progression out of the untreated infected class reduces the average duration of untreated infection in this simplified transmission structure.

Sensitivity interpretation

Positive sensitivity means that increasing the parameter increases $\mathcal{R}_0$. Negative sensitivity means that increasing the parameter decreases $\mathcal{R}_0$. The dashboard uses this idea to rank parameters and interventions according to their effect on HIV transmission potential.

Chapter 5

Numerical Scheme and Dashboard

5.1 Fractional Numerical Scheme

The fractional-order model is solved numerically using the fractional Adams–Bashforth–Moulton predictor–corrector method (Diethelm et al., 2002; Diethelm, 2010). This method is widely used for Caputo fractional differential equations because it approximates the Volterra integral representation of the system and naturally accounts for memory effects.

Consider the general Caputo fractional initial value problem

$$ {}^{C}D_t^q y(t) = f(t, y(t)), \qquad y(0) = y_0, \qquad 0 < q \leq 1. \tag{5.1} $$

Its equivalent integral form is

$$ y(t) = y_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} f(s, y(s))ds. \tag{5.2} $$

Let

$$ t_n = nh, \qquad n = 0, 1, \ldots, M, $$

where $h > 0$ is the time step.

5.1.1 Predictor Step

The Adams–Bashforth predictor estimates y_{n+1} using previously computed values:

$$ y_{n+1}^P = y_0 + \frac{h^q}{\Gamma(q+1)} \sum_{j=0}^{n} b_{j,n+1} f(t_j, y_j), \tag{5.3} $$

where

$$ b_{j,n+1} = (n+1-j)^q - (n-j)^q, \qquad j = 0, 1, \ldots, n. \tag{5.4} $$

5.1.2 Corrector Step

The Adams–Moulton corrector improves the predicted value by evaluating the function at t_{n+1}:

$$y_{n+1} = y_0 + \frac{h^q}{\Gamma(q+2)} \left[f(t_{n+1}, y_{n+1}^P) + \sum_{j=0}^{n} a_{j,n+1} f(t_j, y_j) \right]. \tag{5.5}$$

The corrector weights are

$$a_{j,n+1} = \begin{cases} n^{q+1} - (n-q)(n+1)^q, & j = 0, \\ (n-j+2)^{q+1} + (n-j)^{q+1} - 2(n-j+1)^{q+1}, & 1 \le j \le n, \\ 1, & j = n+1. \end{cases} \tag{5.6}$$

Corrected ABM weight formula

The first corrector coefficient is

$$a_{0,n+1} = n^{q+1} - (n-q)(n+1)^q.$$

This is the standard form used in the Diethelm fractional Adams–Bashforth–Moulton predictor–corrector method. It replaces the earlier indexing form that may lead to incorrect weights.

5.2 Application of the Scheme to the SITA System

For the SITA system, define

$$Y_n = (S_n, I_n, T_n, A_n)^T.$$

The right-hand side vector is

$$F(t_n, Y_n) = \begin{pmatrix} \Lambda - \lambda_n S_n - \mu S_n \\ \lambda_n S_n - (\tau(1+u_3) + \delta + \mu) I_n \\ \tau(1+u_3) I_n - (\rho(1-u_4) + \mu) T_n \\ \delta I_n + \rho(1-u_4) T_n - (\mu + d) A_n \end{pmatrix},$$

where

$$\lambda_n = \frac{\beta_0 (1-u_1)(1-u_2)(I_n + \eta T_n)}{S_n + I_n + T_n + A_n}.$$

The vector predictor is

$$Y_{n+1}^{P} = Y_0 + \frac{h^q}{\Gamma(q+1)} \sum_{j=0}^{n} b_{j,n+1} F(t_j, Y_j), \tag{5.7}$$

and the vector corrector is

$$Y_{n+1} = Y_0 + \frac{h^q}{\Gamma(q+2)} \left[F(t_{n+1}, Y_{n+1}^{P}) + \sum_{j=0}^{n} a_{j,n+1} F(t_j, Y_j) \right]. \tag{5.8}$$

After each numerical step, nonnegative projection is applied:

$$Y_{n+1} \leftarrow \max(Y_{n+1}, 0),$$

component by component. This prevents small numerical round-off errors from producing negative population values.

5.3 Numerical Algorithm

Algorithm: Fractional ABM Simulation for the SITA Model

1. Set the initial vector
$$Y_0 = (S_0, I_0, T_0, A_0)^T.$$
2. Select model parameters and intervention levels.
3. Choose the fractional order q, final time T_f, and step size h.
4. Generate the time grid $t_n = nh$, $n = 0, 1, \ldots, M$.
5. For each time step $n = 0, 1, \ldots, M-1$:
   (a) Compute predictor weights $b_{j,n+1}$.
   (b) Evaluate the predictor Y_{n+1}^{P}.
   (c) Compute corrector weights $a_{j,n+1}$.
   (d) Evaluate the corrected value Y_{n+1}.
   (e) Apply nonnegative projection if necessary.
6. Compute $\mathcal{R}_0$, peak infection, final states, and summary indicators.
7. Send results to the dashboard for visualisation.

5.4 Numerical Accuracy and Consistency Checks

The numerical method is checked using the following consistency principles:

1. **Non-negativity**: all simulated compartments must remain nonnegative.

2. **Boundedness**: the total population $N(t)$ should remain below the finite-time envelope $\max\{N(0), \Lambda/\mu\}$ and approach the long-term bound Λ/μ.

3. **Classical limit**: when $q = 1$, the fractional model should behave like the corresponding ordinary differential equation model.

4. **Step-size reliability**: reducing the step size h should not change the qualitative conclusions of the simulation.

5. **Threshold consistency**: scenarios with $\mathcal{R}_0 < 1$ should generally show declining infection, while scenarios with $\mathcal{R}_0 > 1$ may show persistence.

Table 5.1: Step-size reliability check for the baseline simulation.

Step size h	Time steps	Final I	Final T	Final A
0.20	251	3.1127	94.2720	3.5882
0.10	501	3.1194	94.2676	3.5881
0.05	1001	3.1194	94.2676	3.5880

Table 5.1 shows that reducing the step size from $h = 0.20$ to $h = 0.05$ changes the final infected population by less than 0.01 individuals in the baseline simulation. This supports the numerical reliability of the qualitative conclusions used in Chapter 6.

5.5 Simulation Scenarios

Eight simulation scenarios are used to examine the effects of social behaviour interventions and fractional memory. The scenarios are designed for comparative mathematical analysis rather than direct clinical prediction.

Table 5.2: Simulation scenarios used for intervention and memory comparison.

No.	Scenario	u_1	u_2	u_3	u_4	q	Purpose
1	No intervention	0.0	0.0	0.0	0.0	0.95	Baseline fractional trajectory without behavioural control.
2	Awareness only	0.7	0.0	0.0	0.0	0.95	Isolates the effect of awareness and education.
3	Safer behaviour only	0.0	0.7	0.0	0.0	0.95	Isolates the effect of condom use and safer behaviour.
4	Testing only	0.0	0.0	0.7	0.0	0.95	Tests faster movement from infection to treatment.
5	Adherence only	0.0	0.0	0.0	0.8	0.95	Tests reduced progression from treatment to AIDS.
6	Combined intervention	0.7	0.7	0.6	0.8	0.95	Represents integrated prevention, testing, treatment, and adherence support.
7	Ordinary model	0.5	0.5	0.5	0.5	1.00	Removes fractional memory for comparison with the classical model.
8	High memory effect	0.4	0.4	0.4	0.4	0.75	Tests stronger memory effects on the epidemic trajectory.

Scenario interpretation

Scenarios 1–5 identify individual intervention effects. Scenario 6 tests a combined strategy. Scenarios 7 and 8 isolate the impact of the fractional order by comparing the ordinary model with a stronger-memory fractional model.

5.6 Fractional-Order Comparison

To study memory effects, the system is simulated using

$$q = 1.00, \qquad q = 0.95, \qquad q = 0.85, \qquad q = 0.75.$$

When $q = 1$, the model reduces to the classical first-order system. When $q < 1$, memory effects are introduced through the Caputo kernel. Smaller values of q represent stronger memory and slower decay of past influence.

The dashboard compares the infected curve $I(t)$, the treated curve $T(t)$, the AIDS-stage curve $A(t)$, and the I–T phase trajectory under different values of q.

5.7 Dashboard Architecture

The dashboard is implemented as a small scientific web system with five layers:

Layer	Function
Model layer	Stores the fractional SITA equations and intervention functions.
Solver layer	Runs the fractional ABM predictor–corrector numerical method.
Analysis layer	Computes $\mathcal{R}_0$, final states, peak infection, and sensitivity indices.
API layer	Provides simulation, scenario, memory comparison, sensitivity, and export endpoints.
Interface layer	Displays sliders, cards, graphs, tables, and downloadable outputs.

5.8 Dashboard Features

The dashboard contains the following major features:

- Fractional order slider for q.
- Parameter sliders for biological and intervention parameters.
- Real-time $\mathcal{R}_0$ computation.
- Time-series graphs for $S(t)$, $I(t)$, $T(t)$, $A(t)$, and $N(t)$.
- Scenario comparison module.
- Memory-effect comparison for different values of q.
- Sensitivity analysis bar chart.
- Dashboard screenshot and result export functionality.
- CSV export for simulation outputs.

5.9 System Testing and Validation

The dashboard implementation was checked using automated unit and API tests. The test suite covers the reproduction-number formula, intervention effects, right-hand-side model function, nonnegative numerical output, input validation, scenario comparison, sensitivity analysis, memory comparison, and CSV export. The command

```
python -m pytest tests -p no:cacheprovider -q
```

returned

34 passed

on the project code used for this report.

Dashboard validation summary

The tests confirm that the model equations execute without shape errors, the ABM solver returns nonnegative compartment values, invalid user inputs are rejected, $\mathcal{R}_0$ is computed consistently with the analytical formula, and the main dashboard API endpoints return usable simulation, scenario, sensitivity, memory, and export results.

Chapter 6

Results and Discussion

6.1 Baseline Fractional Simulation

This section presents the baseline numerical simulation of the fractional-order SITA model using the parameter values and initial conditions described in the numerical scheme and dashboard chapter. The baseline simulation was performed for 50 years with step size $h = 0.2$ and fractional order $q = 0.95$. The initial population was

$$S(0) = 10000, \qquad I(0) = 150, \qquad T(0) = 80, \qquad A(0) = 20.$$

The social behaviour intervention levels used in the baseline dashboard configuration were

$$u_1 = 0.40, \qquad u_2 = 0.50, \qquad u_3 = 0.60, \qquad u_4 = 0.70.$$

These values represent moderate-to-strong awareness, safer behaviour, testing and treatment-seeking, and adherence support.

Table 6.1: Baseline parameter values used in the dashboard simulation.

Parameter	Baseline value	Source status and interpretation
Λ	100	Normalized recruitment value used to scale the academic simulation.
μ	0.02	Demographic mortality value used to define the long-term population envelope.
β_0	0.30	Baseline transmission intensity guided by HIV compartmental modelling studies.
η	0.10	Reduced infectiousness under treatment, reflecting ART-related transmission reduction.
τ	0.20	Treatment initiation rate used before testing-intervention adjustment.
δ	0.10	Progression rate from untreated infection to AIDS-stage compartment.
ρ	0.03	Progression rate from treated class to AIDS before adherence intervention.
d	0.33	AIDS-induced mortality parameter used in HIV/AIDS model simulations.
q	0.95	Fractional order selected to represent moderate memory effects.
u_1, u_2, u_3, u_4	0.40, 0.50, 0.60, 0.70	User-defined intervention levels for awareness, safer behaviour, testing, and adherence.

These baseline values are literature-informed and scaled for qualitative mathematical simulation and dashboard demonstration (Anderson and May, 1991; Silva and Torres, 2017; Günerhan et al., 2020; Unaegbu et al., 2021; RBC, 2020). They are not claimed to be fitted estimates from patient-level Rwanda data.

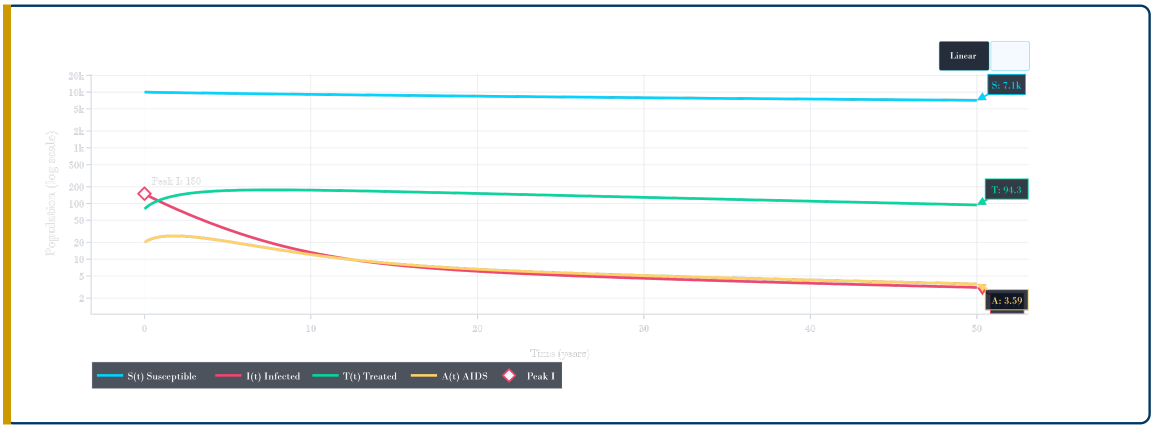
Under this configuration, the effective rates were

$$\beta_{\text{eff}} = 0.0900, \qquad \tau_{\text{eff}} = 0.3200, \qquad \rho_{\text{eff}} = 0.0090.$$

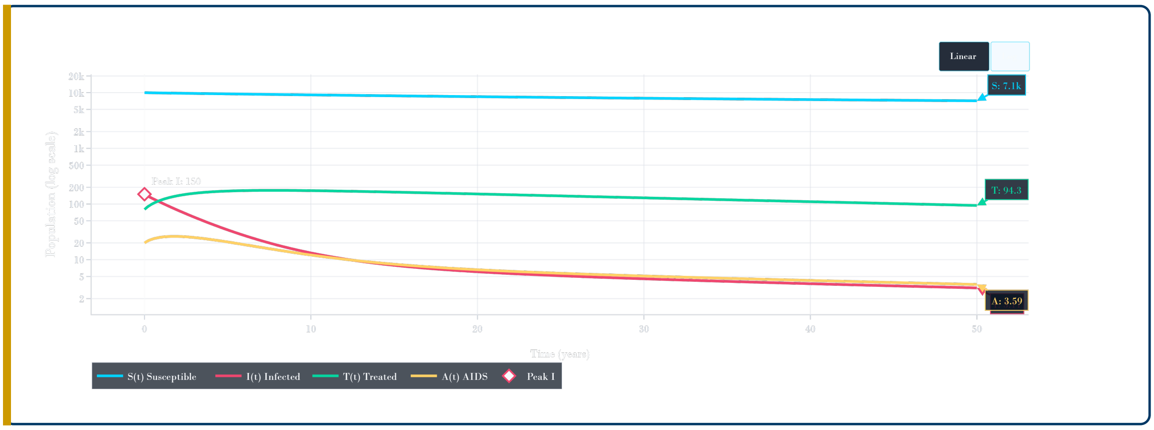
The computed basic reproduction number was

$$\mathcal{R}_0 = 0.430,$$

which is less than one. This indicates that, under the selected intervention-adjusted parameter values, the disease-free equilibrium is locally stable in the threshold sense and the simulated infection trajectory declines over time.



(a) Baseline SITA time-series graph in the dashboard linear view.



(b) Baseline SITA time-series graph in logarithmic view.

Figure 6.1: Baseline fractional-order SITA simulation for $q = 0.95$ showing $S(t)$, $I(t)$, $T(t)$, and $A(t)$.

Figure 6.1 shows the baseline behaviour of the four model compartments. The susceptible population $S(t)$ remains the largest compartment throughout the simulation and decreases gradually from the initial value of 10000 to approximately 7146.82 at the final time. The infected population $I(t)$ starts at its highest value of 150 and then decreases continuously, reaching approximately 3.11 by year 50. This decline is consistent with the computed value $\mathcal{R}_0 = 0.430 < 1$, which indicates controlled transmission under the selected intervention levels.

The treated class $T(t)$ initially increases because infected individuals are moved into treatment through the testing and treatment-seeking intervention. After reaching a higher level, it gradually decreases and ends at approximately 94.27. The AIDS-stage class $A(t)$ remains low compared with the susceptible and treated compartments and ends at approximately 3.59. The logarithmic view is especially useful because it makes the smaller infected, treated, and AIDS-stage curves visible on the same graph as the much larger susceptible population.

Table 6.2: Baseline simulation summary for the fractional SITA model.

Quantity	Value	Interpretation
$\mathcal{R}_0$	0.430	Controlled epidemic status
Peak infected	150.00	Occurs at $t = 0.0$ years
Final susceptible, $S(50)$	7146.82	Susceptible population at final time
Final infected, $I(50)$	3.11	Infected population at final time
Final treated, $T(50)$	94.27	Treated population at final time
Final AIDS, $A(50)$	3.59	AIDS-stage population at final time

The baseline results show a strong decline in the infected class. The infected population starts at 150 individuals and decreases to approximately 3.11 individuals by the end of the simulation period. The treated class remains higher than the infected and AIDS classes at the final time, which is consistent with the effect of testing and treatment-seeking intervention. The AIDS-stage population also remains low, which reflects the effect of adherence support in reducing progression from treatment to AIDS.

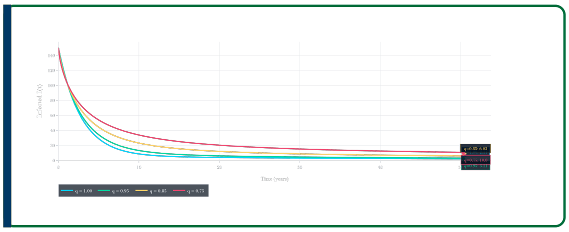
6.2 Effect of Fractional Order

The fractional order q controls the strength of memory in the Caputo fractional derivative. When $q = 1$, the model reduces to the classical ordinary differential equation model. When $0 < q < 1$, the model contains memory effects, meaning that previous states of the system continue to influence the present dynamics through the fractional kernel.

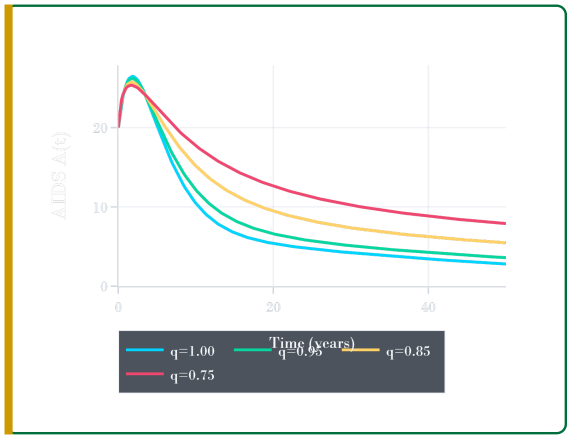
To examine the role of memory, the model was simulated using four values of the fractional order:

$$q = 1.00, \qquad q = 0.95, \qquad q = 0.85, \qquad q = 0.75.$$

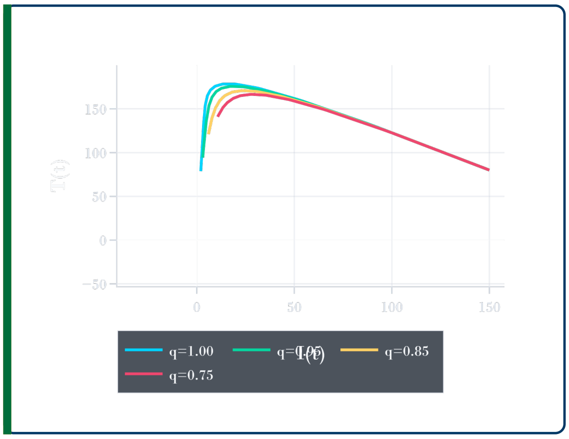
All other parameter values were kept fixed so that the differences in the trajectories could be attributed to the change in q.



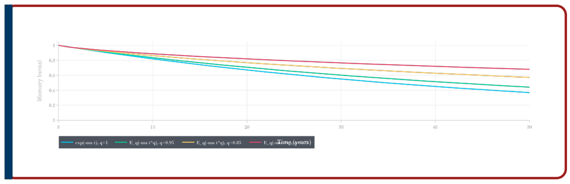
(a) Infected population $I(t)$.



(b) AIDS-stage population $A(t)$.



(c) Phase trajectory in the I–T plane.



(d) Mittag–Leffler memory kernel.

Figure 6.2: Two-by-two memory-effect dashboard visualisation for $q = 1.00$, $q = 0.95$, $q = 0.85$, and $q = 0.75$.

Figure 6.2 compares the model response for $q = 1.00$, $q = 0.95$, $q = 0.85$, and $q = 0.75$ in a compact two-by-two layout. In Figure 6.2a, all infected curves begin from the same initial value, but their decline rates are different. The ordinary model with $q = 1.00$ decreases fastest, while the lower fractional order $q = 0.75$ decreases slowest and remains highest at the final time. This confirms that stronger fractional memory delays the movement of the system toward the disease-free state.

In Figure 6.2b, the AIDS-stage population first increases slightly and then decreases over time. The curve for the lower fractional order $q = 0.75$ remains higher for longer, while the ordinary case $q = 1.00$ decreases faster. This means that stronger memory effects can delay the reduction of the AIDS-stage population.

The phase graph in Figure 6.2c shows the relationship between the infected and treated compartments. The trajectories move through a similar region of the phase plane, but the curves separate according to the fractional order. This indicates that changing q

affects not only the size of each compartment but also the dynamic relationship between infection and treatment.

Figure 6.2d shows the fractional memory kernel. The ordinary exponential case decreases faster, while the Mittag–Leffler type curves associated with fractional orders decay more slowly. This supports the interpretation that fractional models preserve past effects for a longer period than classical ordinary differential equation models.

Table 6.3: Comparison of simulation outcomes for different fractional orders.

q	Peak I	Time of peak	Final I	Final T	Final A
1.00	150.00	0.00	2.11	78.78	2.79
0.95	150.00	0.00	3.11	94.27	3.59
0.85	150.00	0.00	6.03	120.99	5.47
0.75	150.00	0.00	10.79	140.94	7.90

The results show that decreasing the fractional order changes the long-term behaviour of the system. For $q = 1.00$, the final infected population is approximately 2.11, while for $q = 0.75$ it increases to approximately 10.79. Similarly, the treated and AIDS-stage populations are higher for smaller values of q. This indicates that stronger memory effects slow the rate at which the system approaches the controlled state. In epidemiological terms, this supports the idea that past behavioural patterns, treatment history, and adherence behaviour may continue to influence HIV dynamics over time.

6.3 Effect of Single Interventions

This section examines the effect of each intervention when applied separately. The purpose is to identify how awareness, safer sexual behaviour, testing and treatment-seeking, and adherence influence the model before they are combined. Four single-intervention scenarios were considered:

awareness only, safer behaviour only, testing only, adherence only.

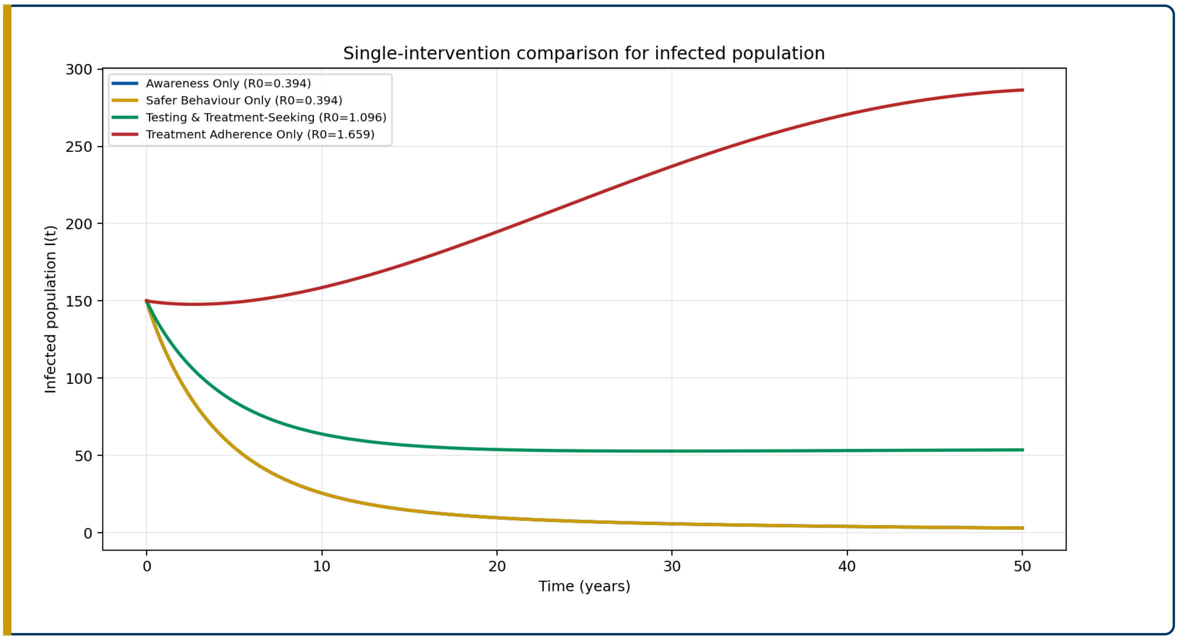


Figure 6.3: Effect of single interventions on the infected population.

Table 6.4: Single-intervention scenario results.

Scenario	$\mathcal{R}_0$	Peak I	Final I	Final A
Awareness only	0.394	150.00	3.03	5.24
Safer behaviour only	0.394	150.00	3.03	5.24
Testing and treatment-seeking	1.096	150.00	53.60	43.90
Treatment adherence only	1.659	286.44	286.44	99.45

The awareness-only and safer-behaviour-only scenarios both reduce the effective transmission rate and give $\mathcal{R}_0 = 0.394$, which is below the epidemic threshold. This shows that interventions acting directly on transmission can strongly reduce the reproduction number. Testing and treatment-seeking alone improves movement from the infected class to the treated class, but it does not directly reduce transmission in the same way as awareness or safer behaviour. As a result, the reproduction number remains above one in the testing-only case. Adherence support alone reduces progression from treatment to AIDS, but it does not sufficiently reduce new infections when used in isolation; therefore, the model still shows persistence under that scenario.

These results show that single interventions can be useful, but their effectiveness depends on which model pathway they influence. Interventions that reduce transmission have a stronger direct effect on $\mathcal{R}_0$, while treatment-related interventions mainly influence the distribution between infected, treated, and AIDS-stage compartments.

6.4 Combined Intervention Strategy

The combined intervention strategy applies several forms of control at the same time. This is important because HIV transmission is not controlled by one factor only. Awareness, safer behaviour, testing, treatment access, and adherence all contribute to the long-term epidemic outcome.

Three combined scenarios were simulated: a moderate combined intervention, the main combined intervention, and a strong combined intervention. The results are shown in Table 6.5.

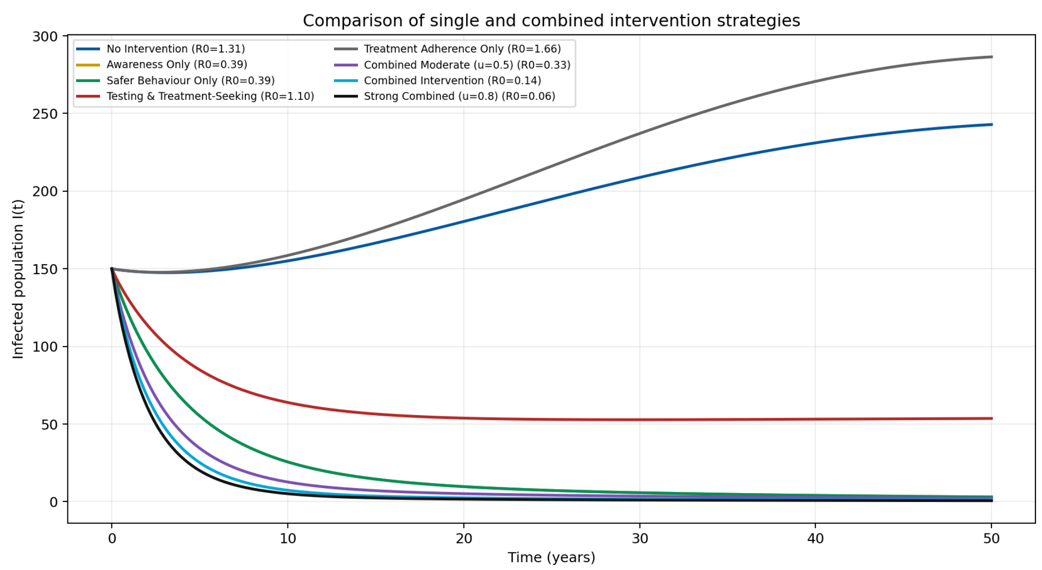


Figure 6.4: Comparison of single and combined intervention strategies.

Table 6.5: Combined-intervention scenario results.

Scenario	$\mathcal{R}_0$	Peak I	Final I	Final A
Combined intervention	0.137	150.00	1.03	1.80
Combined moderate ($u = 0.5$)	0.332	150.00	2.21	3.95
Strong combined ($u = 0.8$)	0.060	150.00	0.64	1.61

The combined intervention strategies produce the strongest control results. The main combined intervention gives $\mathcal{R}_0 = 0.137$, while the strong combined intervention gives $\mathcal{R}_0 = 0.060$. Both values are far below one, indicating strong epidemic control. The final infected population also decreases substantially, reaching approximately 1.03 under the main combined intervention and 0.64 under the strong combined intervention.

These results support the conclusion that combined interventions are more effective than isolated intervention strategies. In particular, reducing transmission while also improving testing, treatment uptake, and adherence produces better long-term control than focusing on one pathway alone.

6.5 Sensitivity Analysis Results

Sensitivity analysis was used to identify the parameters and intervention controls that have the strongest influence on the basic reproduction number. The normalized sensitivity index measures the proportional change in $\mathcal{R}_0$ caused by a proportional change in a parameter. A positive sensitivity index means that increasing the parameter increases $\mathcal{R}_0$, while a negative index means that increasing the parameter reduces $\mathcal{R}_0$.

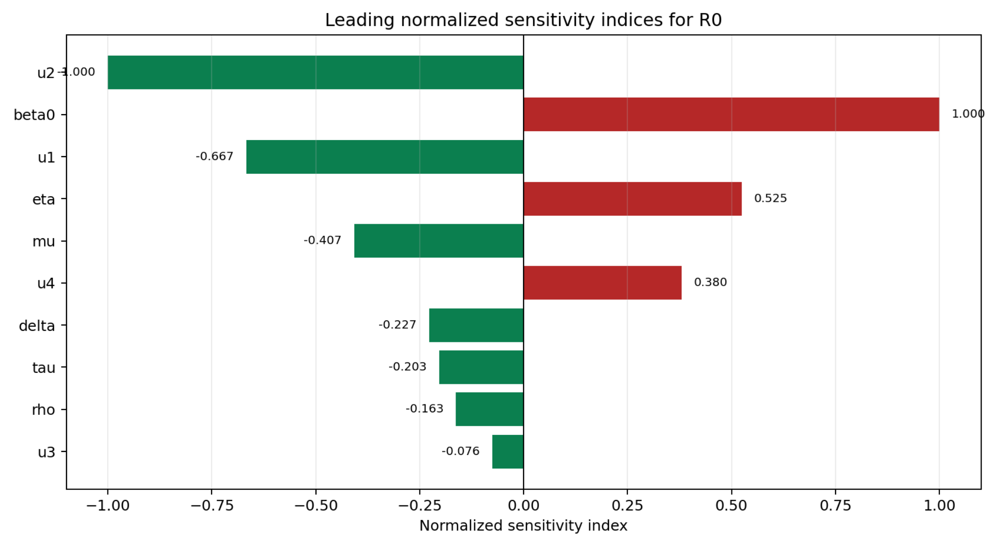


Figure 6.5: Normalized sensitivity indices for $\mathcal{R}_0$.

Table 6.6: Leading normalized sensitivity indices for $\mathcal{R}_0$.

Parameter or control	Sensitivity index
u_2	-1.000
β_0	1.000
u_1	-0.667
η	0.525
μ	-0.407
u_4	0.380
δ	-0.227
τ	-0.203

The baseline transmission rate β_0 has a positive sensitivity index of approximately 1.000, which means that increasing the baseline transmission rate increases $\mathcal{R}_0$ almost proportionally. The safer-behaviour intervention u_2 has a sensitivity index of approximately -1.000, showing that increasing safer behaviour strongly reduces $\mathcal{R}_0$. Awareness u_1 also has a negative sensitivity index, confirming its role in reducing effective transmission.

The sensitivity results therefore support the intervention findings: reducing transmission is central to controlling the epidemic threshold. Treatment and adherence parameters also affect the system, especially by changing movement into treatment and progression to AIDS, but transmission-related factors are the strongest drivers of $\mathcal{R}_0$ in this configuration.

6.6 Dashboard Demonstration

The computational part of this report was implemented as an interactive Flask dashboard. The dashboard allows the user to adjust initial conditions, biological parameters, fractional order, and intervention levels. It then displays time-series graphs, $\mathcal{R}_0$, stability interpretation, scenario comparison, memory-effect comparison, sensitivity analysis, and export tools.

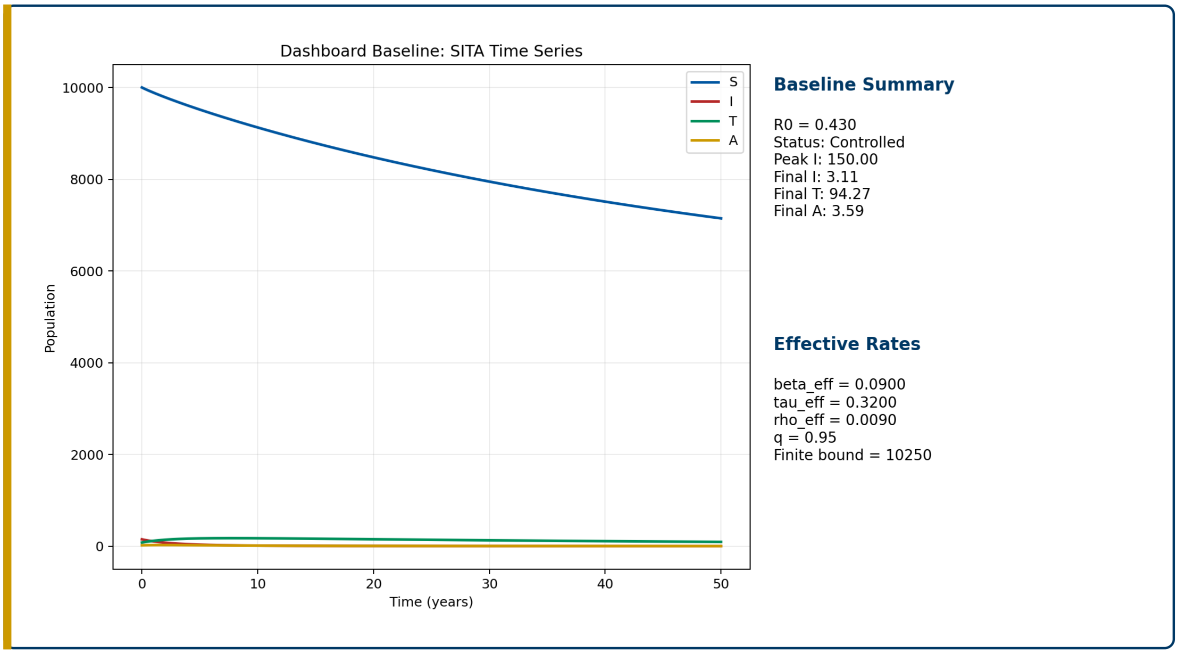


Figure 6.6: Dashboard baseline simulation interface.

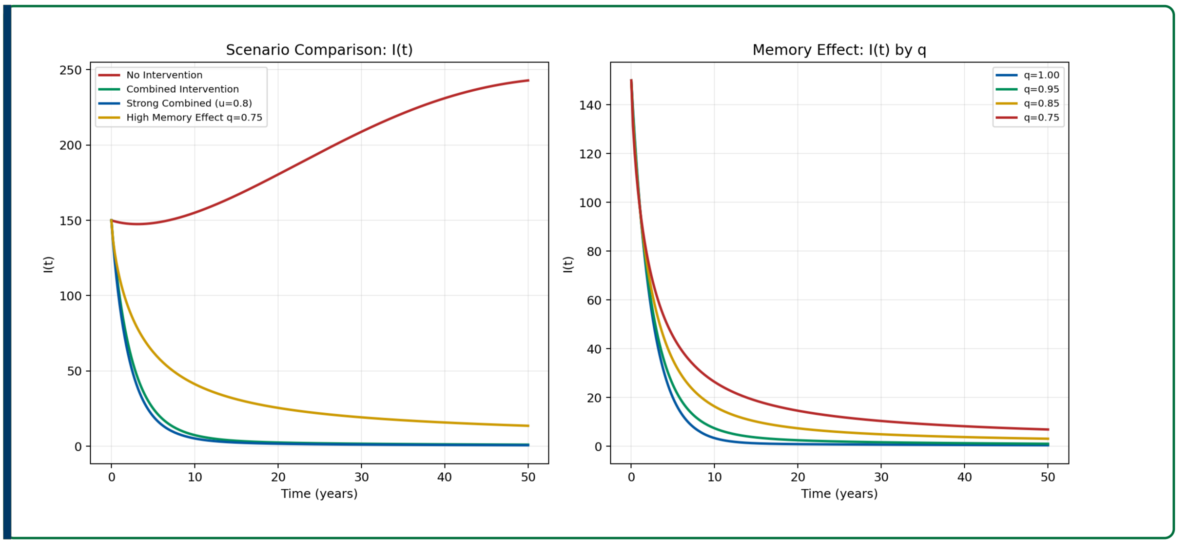
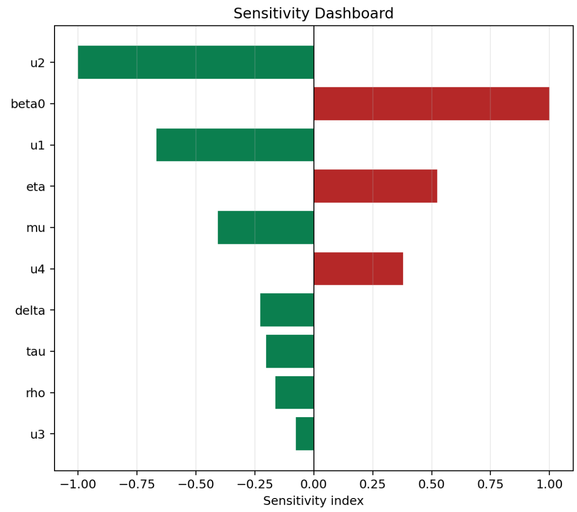


Figure 6.7: Dashboard scenario and memory-effect visualisation.



Chapter 6 Result Generator: Baseline Table

Quantity	Value	Interpretation
R0	0.430	Controlled
Peak infected	150.000	at t=0.0 years
Final susceptible	7146.818	S(t_end)
Final infected	3.113	I(t_end)
Final treated	94.272	T(t_end)
Final AIDS	3.588	A(t_end)

Figure 6.8: Dashboard sensitivity analysis and report-result generator.

The dashboard supports the report in three main ways. First, it provides visual evidence for the analytical model by plotting the numerical trajectories of the four compartments. Second, it allows intervention scenarios to be compared using the same baseline parameter values. Third, it provides export tools for CSV tables, figures, and report-ready interpretation text. This makes the mathematical model easier to communicate during project writing and presentation.

6.7 Public Health Interpretation

The numerical results have direct public-health interpretation, although the model is intended for academic simulation rather than clinical prediction. The results show that interventions reducing effective transmission have a strong effect on the reproduction number. In practical terms, awareness campaigns, risk-reduction education, condom use, and safer sexual behaviour can reduce the probability that susceptible individuals become infected.

Testing and treatment-seeking interventions are also important because they move infected individuals into treatment more quickly. This reduces the untreated infected population and increases the treated population. Treatment adherence reduces progression from treatment to AIDS, helping to keep the AIDS-stage population lower. These findings are consistent with the idea that HIV control requires both prevention and treatment support.

In the context of Rwanda's HIV response, the model supports the importance of integrated intervention programmes. Rwanda has made progress in HIV testing, treatment, and viral suppression, but continued prevention and adherence support remain important. The combined-intervention simulations show that applying behavioural and biomedical interventions together produces stronger control than applying one intervention alone.

The fractional-order results also provide an important interpretation. When $q < 1$, past behaviour and treatment history influence present dynamics through memory effects. This reflects the fact that public health interventions may not act instantaneously; instead, their influence may persist over time through community awareness, adherence habits, stigma reduction, and accumulated treatment experience.

Chapter 7

Conclusion and Recommendations

7.1 Conclusion

This report develops a fractional-order SITA HIV transmission model incorporating social behaviour interventions. The use of Caputo fractional derivatives allows the model to capture memory effects, while the Gamma and Mittag-Leffler functions provide the mathematical foundation for the fractional framework. The model connects HIV transmission, treatment initiation, AIDS progression, and behavioural interventions in a single mathematical structure.

The theoretical analysis focuses on positivity, boundedness, disease-free equilibrium, the basic reproduction number, and fractional-order stability. The computational component demonstrates how fractional dynamics and social behaviour interventions can be explored through simulation and dashboard visualisation.

7.2 Recommendations

1. Future work may include age or gender structure.
2. Future studies may estimate parameters using Rwanda-specific data.
3. Stochastic fractional models may be considered for small populations.
4. The dashboard may be extended into a decision-support educational tool.
5. Other fractional operators, such as Caputo–Fabrizio or Atangana–Baleanu derivatives, may be compared with the Caputo derivative.

7.3 Limitations of the Study

1. The model assumes homogeneous mixing.
2. Parameter values are literature-informed and scaled for scenario comparison, but they are not fitted to patient-level Rwanda data.
3. The model is deterministic and does not include random effects.
4. The intervention functions simplify complex human behaviour into bounded mathematical controls.
5. The dashboard is for academic simulation, not clinical prediction.

Chapter A

Suggested Project Folder Structure

```
fractional_hiv_project_dashboard/
|
|-- app.py
|-- config.py
|-- requirements.txt
|-- README.md
|
|-- model/
|    |-- fractional_sita_model.py
|    |-- fractional_solver.py
|    |-- reproduction_number.py
|    |-- scenarios.py
|    |-- sensitivity.py
|    `-- validation.py
|
|-- routes/
|    |-- simulation_routes.py
|    |-- export_routes.py
|    `-- health_routes.py
|
|-- templates/
|    |-- base.html
|    |-- index.html
|    |-- dashboard.html
|    |-- documentation.html
|    `-- about.html
|
|-- static/
|    |-- css/
|    |-- js/
|    |-- assets/
|    `-- vendor/
```

```
|
|-- data/
|   |-- baseline_parameters.json
|   |-- scenario_presets.json
|   `-- sample_outputs.csv
|
`-- outputs/
    |-- csv/
    |-- figures/
    `-- reports/
```

Chapter B

Fractional ABM Solver Skeleton

```python
import numpy as np
from scipy.special import gamma


def fractional_abm_solver(rhs, y0, t, q, params):
    """Predictor-corrector solver for Caputo fractional systems."""
    n = len(t)
    h = t[1] - t[0]
    y = np.zeros((n, len(y0)), dtype=float)
    y[0] = np.maximum(y0, 0.0)
    fvals = np.zeros_like(y)
    fvals[0] = rhs(t[0], y[0], params)

    for k in range(1, n):
        history = np.arange(k, 0, -1, dtype=float)
        b = history**q - (history - 1.0)**q
        y_pred = y[0] + (h**q / gamma(q + 1.0)) * (b @ fvals[:k])

        a = np.empty(k, dtype=float)
        a[0] = (k - 1)**(q + 1) - (k - 1 - q) * k**q
        if k > 1:
            m = np.arange(k - 1, 0, -1, dtype=float)
            a[1:] = (m + 1.0)**(q + 1) + (m - 1.0)**(q + 1) - 2.0*m**(q + 1)

        y[k] = y[0] + (h**q / gamma(q + 2.0)) * (
            rhs(t[k], y_pred, params) + (a @ fvals[:k])
        )
        y[k] = np.maximum(y[k], 0.0)
        fvals[k] = rhs(t[k], y[k], params)
```

```
    return y
```

Chapter C

Proposed Baseline Parameter Table

Parameter	Description	Baseline Status
β_0	Baseline transmission rate	Literature-guided
μ	Natural death rate	Demographic context
τ	Treatment initiation rate	Literature-guided
δ	Progression from infected to AIDS	Literature-guided
ρ	Progression from treated to AIDS	Literature-guided
d	AIDS-induced mortality	Literature-guided
η	Relative infectiousness under treatment	Literature-guided
q	Fractional order	Simulation parameter
u_1, u_2, u_3, u_4	Social behaviour interventions	User-defined

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